# Using BERT to extract emotions from personal journals

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***Abstract*—Keeping a journal can prove highly beneficial when it comes to confidence and overcoming psychological issues. There are plenty of applications that aim to help treat depression with the help of machine learning, using chat interfaces. However, little attention has been given to digital journals which take advantage of the latest AI technologies to help users better understand their emotions and track their evolution, for therapeutic purposes. A good understanding of human emotions is essential in the context of identifying, treating, and keeping track of psychological issues. There is extensive research in terms of emotion detection through video, audio, or pictures, and less when it comes to text input, despite the amount of data available publicly. We propose an application that, based on a model specialized in emotion detection, provides the user with meaningful insights which can be used to make informed decisions and take action.**

***Index Terms*—BERT, emotion detection, machine learning**

## I. Introduction

There are multiple ways in which people exhibit behavior based on their emotions. The research done on detecting emotions from videos and images has been rather comprehensive, whereas text-based systems are still considerably less popular [1]. This is partly because of the harder patterns to identify and more complicated expressions. Moreover, a significant number of adolescents suffer from depression partly due to high levels of stress or behavioral inhibition [2]. Keeping a journal can create a sense of belonging and help develop an identity [3] which are basic needs, necessary for sanity and mental health. Taking this into consideration, a digital journal that uses machine learning for emotion detection and assists users in understanding their inner thoughts would benefit society greatly. Further more, the process can be applied later on to the messages written on different social platforms, establishing the state of mind of the person.

Advancements in psychology provide personalized treatments based on dialogue, but they require a key component: the person has to go to access these medical services. Most of the time a person does not realize his psychological condition or does not want to accept it, even thought there are friends and family members who encourage him to seek help. Because of misconceptions, social norms, and the shame to call for psychiatric help, some individuals are reluctant to reveal certain details of their inner emotions or to go to a doctor. A journal takes away the interaction component and allows the free expression of thoughts, thus being a better method to evaluate one's state of mind. Applications that aim to help with psychological issues on the basis of emotional understanding, such as Tess, have the disadvantage of requiring direct interaction with an entity. As further motivation, writing in a diary has other benefits: it helps improve writing skills in an efficient manner [4] and increases one's confidence due to expressing emotions.

Given the ascension in terms of popularity of the social media platforms, on which people consistently express their perceptions and feelings, and the fact that a large portion of that information is in text format, we believe that people are already accustomed to expressing emotions and inner thoughts through this means. Even if the people are used to express their emotions on social media, due to its public character (many people can see a post), most of the people don't express their negative feelings, showing just the good parts and the accomplishments in their lives. Using a personal smart journal which helps them understand their issues and monitor their progress is a highly beneficial aspect. Furthermore, the journal would help eliminate the risk of bullying which is associated with free speech on social media platforms. In an article published in Schizophrenia Bulletin, there has been research carried out which found that 66.4% of people suffering from psychosis owned a mobile phone [5].

Moreover, keeping a journal can prove to be beneficial in improving writing skills. Chur-Hansen and Anna [4] addressed this issue and suggested that when it comes to writing essays, the teachers often struggle to find the time for assessing the papers for the assignments because of the imbalance between the number of tutors and the number of students. Because of this situation and the fact that students are expected to deliver large papers only towards the end of the semester, they spend most of the academic year without writing. In the paper, the authors stated that a teacher is rarely able to assign more than a paper every other week to the students. A potential solution would be for the attendants of the course to evaluate their work between them, but this also requires the supervision of a tutor. Keeping a journal would be a better alternative, as it would provide them with the opportunity to write as much as they want to.

Writing implies a series of choices that are based on personal beliefs or preferences. When being in the situation of choosing the right words and the appropriate ideas, we can learn more about the way our mind functions and we can better understand our opinions. Joanne Cooper [3] highlighted how keeping a journal can help one discover their identity and track their evolution. Based also on personal experience, the author stated that writing can help the individual be better organized.

Besides motivating the utility of writing in journals, we fine-tuned BERT [6] using an existing data-set to build a model capable of detecting emotions on a text basis, evaluated its performances, and pointed out its limitations. We did so using the CARER data. In an attempt to solve some of the occurring issues, we created a new data-set and trained the model on it, and obtained better results in certain scenarios, such as when dealing with negations. We developed a desktop application that allows users to express their thoughts freely, get predictions based on their input, and access statistics that ease evolution tracking.

In the next section, we will present the state-of-the-art regarding emotion detection, as well as applications aiming to help cure depression. Shortly after, in Proposed Solution, we will document the development of the application, as well as the configuration of the model, while also motivating the decisions taken along the way. In the next section, **Experimental Results**, we will be analyzing the original data-set, presenting evaluation metrics, and will present a comparison to other models. We will continue with a **Discussion** section, in which we aim to test the model on public journals to try to establish a correlation between tragic events and predictions provided by our model. Last but not least, we will sum the paper up with a **Conclusions** section.

## II. Literature Review

### A. Emotion Detection

There is no system of emotions agreed on by everyone. However, most use a finite number of predefined classes. Russel [7] introduced a system in which at every moment, one's emotional state can be represented on a two-dimensional axis consisting of arousal (relaxed or aroused) and valence (pleasant or unpleasant). This representation has potential of better accuracy and a broader coverage of human emotions. However, it is difficult to find the necessary data for training a model under this definition. Another approach in terms of emotion understanding would be appraisal theories, which interpret them rather as an ever-changing state, as opposed to a process. Certain events can influence one's state, and so they can influence an individual's reaction to further events. This would explain why certain individuals have opposed reactions to similar events.

There are a number of models which aim to perform emotion detection on a basis of text, starting from rule-based systems. In [8] we find a review of sentiment analysis and emotion detection models. Approaches like Naive Bayes, feed-forward networks, and keyword spotting can perform well generally in terms of accuracy, but do not take full advantage of correlations between words and don't fully understand expressions. We would need to train a model on a very large and diverse data-set in order to capture the true context of a language. Thus, taking advantage of transfer learning proves to be highly beneficial. For the classification task, the Transformer architecture was revolutionary, as it replaced traditional approaches, based on RNN or convolutions.

Many of the models which aim to extract such information from text input are variations of BERT [6], which was based on the Transformer architecture. Some of the differences between those would be the number of parameters, the architecture and the pre-training tasks. It is possible to maintain state-of-the-art performances using significantly fewer parameters, and this can be useful, especially when running on edge devices. This is worth considering because the model has to physically fit into memory, and also for the reduced training time.

An example of a model which performs emotion detection would be EmoDet2 [9]. It obtained a precision of 69% and a F1 score of 74% using 4 categories: Angry, Happy, Sad and Other. It achieved this using 5 sub-models: a feed-forward, two models based on the BERT architecture, and two LSTM-based components.

Another example is CARER [10] which uses one-dimensional convolutions and pooling, along with dense layers for classification. The model uses the same data-set as we did and obtained an average F1 score of 70%.

Moreover, in the Feeler [11] paper we find a Vector Space Model, a popular information retrieval method based on representing information in a multi-dimensional space and finding similarities between the data records. Using 5 domain classes: anger, disgust, fear, joy and sadness and data from ISEAR, the model achieved an average F1 score of 67% and an accuracy of 67.2%.

### B. Psychological Applications

Tess [12] would be such an application, representing a chat bot that was designed and trained to be a psychological assistant. The author carried out research to prove Tess's ability to improve mental health. He randomly selected 75 students from universities across the USA, which he later split into 3 teams: a control group, which did not receive access to Tess, and was given a link to an electronic book about mental health, and two groups that were exposed to the chat bot. The first one was granted access for 2 weeks, while the other received 4 weeks of trial. The study revealed that the groups exposed to Tess statistically showed less anxiety and depression symptoms, as measured using data from surveys. Russell Fulmer [13] started from the idea that the students with such issues often avoid seeking professional help due to stigma, financial situation or location. An interactive chat bot would be, in his opinion, a far cheaper and more scalable alternative.

On the other hand, Youper [14] is a mobile application with therapeutic purposes that uses artificial intelligence to treat depression and anxiety. Similarly to Tess, it also uses an interface based on interactive chatting and is based on a decision tree architecture. According to the authors of the paper, there are plenty of applications available on the market which accomplish this task, but there is little research carried out to prove the acceptability and performance of those. They selected 5943 users of the application, out of which 4517 accepted to offer their data for this research. The participants were regularly exposed to symptom assessments in order to establish how useful the application is, and showed fewer symptoms of anxiety and depression.

Another example would be Woebot, an automated chatting platform powered by AI that aims to have cognitive behavioural therapeutic effects. Kathleen Kara Fitzpatrick, Alison Darcy and Molly Vierhile [15] carried out a study in which they tried to prove Woebot's feasibility and acceptability in terms of being a therapeutic tool. They used 70 individuals for the experiment, which interacted with the application 12.14 times on average. The authors concluded that they significantly reduced their symptoms of depression over the study period, compared to a control group that did not.

However, we have to mention that such applications have some limitations. One which is worth mentioning is that they do not take into account a person's past experiences when making predictions. In order to provide the ideal treatment tool, we would need data containing the user's trauma, perceptions, inner thoughts. Training a model which takes into account this information is rather difficult, as there are no such public datasets.

## III. The Proposed Solution

We propose a system (illustrated in Fig 1) to perform emotion detection on free-form text and integrate it into a system that tracks individual user emotions over time. On top of that, the system would allow saving pages and navigation through previously saved pages. What is worth mentioning here is that the system would treat the whole text of a page as a single unit of input, and provide predictions for it accordingly.

We downloaded a pre-trained model to take advantage of the already learned patterns (context of English language, expressions etc). The original BERT has a large number of parameters (179M) making it memory intensive. SmallBert [16] achieves similar performances while using less memory, making it suitable for training on less resourceful devices. Starting from SmallBert architecture, we added a dense neural layer on top of it, which has 6 output neurons in our case, for the classification. We experimented with more dense neural layers before the classification, but found no improvement and the effect was that it made the model slightly over-fit. We also added softmax on top of it, in order to be able to interpret the outputs of the network as probabilities, and used the cache functionality from TensorFlow for improved performance. We use Categorical Cross Entropy as loss function, to maximise the probability for the correct emotion, and Adam [17] as optimizer. However, we also tried training on other optimisers, such as Stochastic Gradient Descent, but found no improvement in the accuracy with respect to the validation set. We also added a Dropout layer, for regularization purposes. Experimenting with various learning rates (5e-5, 3e-5, 9e-5), we obtained the best results using 5e-5 initial learning rate. In order to avoid early overfitting, we will be taking advantage of ”warmup-steps” which imply decreasing the impact of some of the first training steps, limiting their updates on the training parameters. By doing this we can avoid an early convergence towards unwanted features and patterns. Last but not least, we will also be using a callback for saving the best model after each epoch, the criteria for selection will be the categorical accuracy for the validation set.

Based on the previously trained model, we developed a feature that plots a graph containing the evolution of the predicted emotions throughout time. It displays a graph containing probabilities for each of the 6 emotions for the pages of the journal, in which dates are ordered chronologically. This is done using all of the previously saved pages of the journal. The purpose of this functionality is to help the user observe patterns, identify potential issues, and take action, if the user considers necessary. We will later use this functionality to test real, published journals, in order to establish if the model detects the change in the author's mental state.

In order to assure quality, we tested the application using unit tests, as well as exploratory manual testing. During this process, we discovered that the model predicts an empty string to be either anger or sadness (the 2 highest predictions). This could be a valid interpretation of silence, but could also be based on chance.

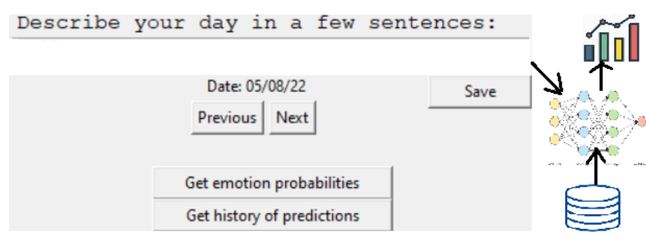


Fig. 1. The overall emotion detection system.

## IV. Experimental Results

### A. The Dataset

We used the dataset gathered by [10], which contains 16k sentences, selected using the Twitter API. The tweets were selected using certain hashtags, later used as ground truths. However, the hashtags had to be removed during preprocessing, to avoid added bias. At a first glance, we can observe that the data is unevenly distributed between classes (Fig. 2) (34.09% joy, 29.66% sadness, 6.45% love, 13.73% anger, 12.31% fear, 3.64% surprise). On average, the sentences contained around 15 words and between 50 and 100 characters (Fig. 3). This uneven class distribution could potentially be an issue, as the model could have a tendency to optimise classification for the most represented classes, considering the optimisation process. We could try to solve this issue by reducing the training records such that all classes have as many records as the one which is least represented, which is *surprise*. Unfortunately, this approach would leave us with very few records left for training. There would be another option, to fill up the training data-set with records such that all classes are equally represented. This also proved to be difficult, as we have over 5000 sentences denoting joy. However, we discovered that the model performed well despite the unbalanced nature of the data-set.

We also trained and tested our model on the 4'th task (subtask A) of SemEval-2017 [18], to establish whether our model is data set biased. We conclude that our model can be extended to other data sets (see Section IV-C).

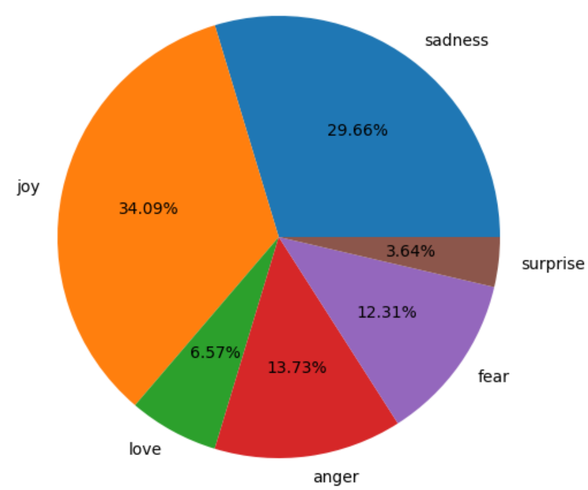


Fig. 2. The distribution of the data set

Another way in which we can look at the data is with respect to the word count. We can observe that most of the records have between 10 and 27 words, with the average being around 15. This could potentially be a limitation for the model, as more context could have been useful for the classification task.

## B. Model Performances

We obtained a maximum accuracy on the CARER testing data set of 92.2%. What is worth mentioning here is that the model tends to mix up anger and fear, respectively love and joy, which come from a similar spectrum (see Table I, II). Another explanation for these misclassifications could be that there are sentences for which the correct classification would not be agreed upon by everyone, or that belong to multiple classes. The testing set contained 10% of all records. However, when trying to test the model on sentences with negated emotions we obtained unwanted results (ex: "i did not have the best day" was given 85% for joy). We deduced that a possible reason for this might be the reduced number of sentences with negations in the training set (272 out of the 16000 contained

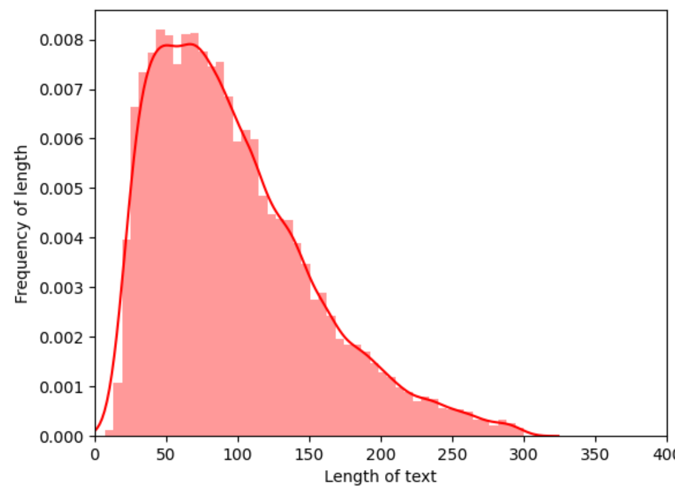


Fig. 3. The distribution of the letter count found in the training set

| | anger | fear | joy | love | sadness | surprise |
|---|---|---|---|---|---|---|
| anger | 256 | 10 | 2 | 0 | 7 | 0 |
| fear | 5 | 202 | 0 | 0 | 7 | 10 |
| joy | 3 | 1 | 655 | 31 | 3 | 2 |
| love | 0 | 0 | 28 | 130 | 1 | 0 |
| sadness | 15 | 6 | 3 | 3 | 554 | 0 |
| surprise | 0 | 11 | 4 | 0 | 3 | 48 |

TABLE I
CONFUSION MATRIX FOR THE MODEL.

the pattern "didnt"). We manually create a data set containing two classes (joy and sadness) and 100 records for each of them, in an attempt to solve this issue. After testing the new model on negations of happiness (10 sentences), we conclude that there is a significant improvement. This could also be caused by the reduced noise, as we tried to focus on expressing emotions when creating the data set. A relevant example in which we could see this difference would be the following : when trying to make predictions on the sentence "i am not ok" using the model trained on the CARER dataset, we obtain 93% probability for joy. On the other hand, the model trained on our custom dataset, manually created, assigned a 80% probability for sadness for the same sentence. Another example could be the following : for "i did not have a good day" the model trained on CARER data assigned a 98% probability for joy, whereas the other model obtained 94% probability for sadness. However, the final model was trained on CARER data.

| | recall | precision | F1 |
|---|---|---|---|
| anger | 0.93 | 0.91 | 0.91 |
| fear | 0.94 | 0.87 | 0.90 |
| joy | 0.94 | 0.94 | 0.94 |
| love | 0.81 | 0.79 | 0.79 |
| sadness | 0.95 | 0.96 | 0.95 |
| surprise | 0.72 | 0.80 | 0.75 |

TABLE II
EVALUATION METRICS.

### C. Comparison to Other Models

Kiran Chowdary posted on Kaggle an attempt at classifying the same dataset (CARER), using a logistic regression approach. He obtained an overall accuracy of 63.35%, by fine-tuning DistilBERT [19], which has 40% fewer parameters than the original BERT model. On the other hand, the model trained by the CARER [10] emotion detection system obtained an average accuracy of 81% using 1D convolutional layers and pooling. We obtained 11% more accuracy than the model developed by the CARER model and 29% more than the Kaggle attempt. We can clearly see the advantage that transfer learning provides in terms of accuracy, when in comparison to models which do not take advantage of pre-learned patterns. Last but not least, we obtained 67.90 % accuracy on the SemEval data set, which would have put us in the first place (comparing the accuracy) for the A sub-task. What is worth mentioning here is that the contest took place before the BERT era, so the participants could not take advantage of this architecture. At Papers with Code we can find the models which performed best on the CARER dataset. Here we find that BERTweet [20] occupies the first poisition, with 94.5% accuracy.

## V. Discussion

We will try to establish a correlation between tragic events occurring in reality and the increased probability of negative emotions, using our model. This will be done using input from published journals, written by people who suffered from tragic events. For each of the journals below, we selected 10 pages before the critical event and 10 after it and used the first sentence or the first two sentences. How will the model interpret these situation changes?

### A. The Diary of Lena Mukhina

Lena was a 16-year-old student in Leningrad (Saint Petersburg). Her journal was written at the same time as Germany invaded the URSS in June 1941, capturing both the last days of peace in the city and the beginning of the war. We are being presented a harsh reality: rationalised food, bombardments and so on. We learn the context of the writings with the help of Valentin Kovalchuk, Aleksandr Rupasov, and Aleksandr Chistikov, who edited the initial diary and composed a foreword containing historical information. According to them, the siege corresponded to a period of starvation in which some even turned to cannibalism. The predictions (Fig. 4, 5) seem to be made according to the events described, as we can observe smaller probabilities for joy and an increase of the other negative emotions. In one of the days we selected for representing the starvation period, we can observe that love received the highest probability. After research, we found out that it was because of her birthday, in which she received a pleasant surprise.


Fig. 4. History of predictions for the period before starvation of Lena Mukhina

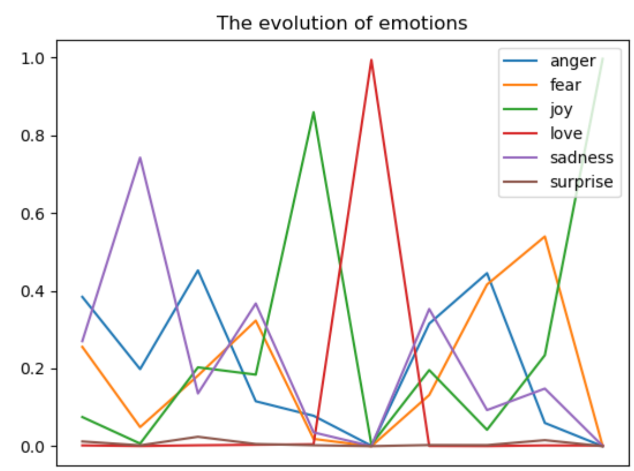


Fig. 5. History of predictions for the period during starvation of Lena Mukhina

### B. Journals: Captain Scott's Last Expedition

In September 1909 the British announced an expedition to the Pole and named Robert Falcon Scott as captain. The ship arrived with great difficulty and was forced to retreat to New Zealand, while the crew was marching towards The South Pole. They failed to be the first team to reach the pole and eventually froze to death in the Antarctic winter. Scott kept writing in his diary as the end approached, and, using our model, we can observe (TableIII) an increase in anger predictions as well as a decline in joy, arguably because of their situation. In the end, they realised that they will not make it, and asked the people back in England to support their families, in return for their sacrifice.

| | anger | fear | joy | love | sadness | surprise |
|---|---|---|---|---|---|---|
| While sailling | 0.13 | 0.17 | 0.29 | 0.00 | 0.34 | 0.04 |
| Before death | 0.38 | 0.11 | 0.13 | 0.00 | 0.35 | 0.00 |

TABLE III
AVERAGE PREDICTED EMOTIONS, WHILE SAILING AND BEFORE THE IMMINENT DEATH OF ROBERT FALCON SCOTT.

## VI. CONCLUSIONS

We trained a model on a emotion detection task and obtained promising results, both on the CARER and SemEval datasets. We also trained the model on a custom data set in an attempt to improve the performances of predictions when dealing with negated sentences. Moreover, we integrated the model in a system capable of detecting emotions and presenting statistics for emotion evolution, in order to help users identify patterns. The last step was to validate it against real life journals in order to establish its correctness, and whether it can identify mood changes based on external events. Based on the testing results, we can conclude that it can identify tragic events, by assigning high probabilities for negative emotions, and lower probabilities for joy.

For future work, we consider developing a model specialized in depression prediction, and integrating it into the system, to consolidate its utility. We also take into consideration developing a model which formulates a question based on the previous pages, and displaying that question as a starting point for the next journal page. For example, if someone would express that they are anxious about an exam, the journal could ask the following: ”Are you still anxious about tomorrow’s test?”. Starting from these models, we aim to build a mobile application with therapeutic purposes which works in the following way: the users may use the application freely, and can choose to send their journal data to a professional psychologist. The psychologist would analyse their emotions and help the users with suggestions, or even schedule a conventional visit, if necessary.

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